Notes: Gentzkow, Shapiro, and Sinkinson (AER, 2011)

The Effect of Newspaper Entry and Exit on Electoral Politics

Contents

1 Big picture

- **Question.** Do local newspapers affect electoral politics? More specifically, how do newspaper entries and exits affect voter turnout, party vote shares, and incumbency advantage?
- **Setting.** The paper builds a new county-level panel of U.S. daily newspapers from 1869 to 2004 and matches it to county-level election outcomes. The main analysis focuses on 1869–1928, when newspapers were the dominant mass political medium.
- **Why this is important.** Media markets are often regulated or discussed as if media supply matters for democracy. But newspaper entry and exit are endogenous: papers enter growing, richer, and politically attractive markets. The paper asks whether the entry or exit itself changes political behavior.
- **Core empirical object.** The main treatment is a change in the number of English-language daily newspapers in a county. The paper observes 3,913 county-years with net newspaper entry and 3,303 county-years with net newspaper exit.
- **Main strategy.** Compare changes in political outcomes in counties that experience newspaper entry or exit to changes in other counties in the same state and year. The specification is in first differences and includes state-year fixed effects and demographic controls.
- **Identification idea.** Entry and exit are discrete events that produce large changes in newspaper availability. The authors exploit the precise timing of these events and use event-study pretrends to diagnose whether omitted economic or political trends are driving the results.
- **Main turnout result.** In 1869–1928, one additional newspaper increases presidential turnout by about **0.3 percentage points**. The estimate is similar for congressional turnout.
- **Market-structure result.** The effect is mainly from the *first* newspaper in a county. The first paper raises turnout by about **1.0 percentage point**; the second and third papers have much smaller and less stable marginal effects.
- **Reader-level interpretation.** The first newspaper raises the share of eligible voters reading at least one paper by about 25 percentage points. This implies that reading a newspaper raises the probability of voting by roughly **4 percentage points**.
- **Time variation.** The effect of newspapers on presidential turnout declines after radio and television become important. In contrast, the effect on congressional turnout remains similar through later periods, consistent with local newspapers remaining important for lower-salience races.
- **Partisan persuasion result.** Partisan newspapers do *not* measurably shift presidential party vote shares. The point estimate for the effect of a Republican paper on the Republican presidential vote share is essentially zero.
- **Incumbency result.** The paper finds no clear evidence that newspapers systematically help or hurt incumbents.
- **Economic takeaway.** Newspapers appear to matter mainly by providing political information that mobilizes voters, rather than by persuading voters to switch parties or systematically changing incumbency advantage.

2 Data and measurement (key objects)

2.1 Newspaper panel

- **Sources.** The newspaper data come from annual newspaper directories:
  - Rowell's *American Newspaper Directory* for 1869–1876;
  - Ayer's *American Newspaper Annual* for 1880–1928;
  - *Editor and Publisher Yearbook* for 1932–2004.
- **Coverage.** The panel includes general-circulation English-language daily newspapers in the United States.
- **Frequency.** The paper uses 1869 and every presidential election year from 1872 to 2004. Since election outcomes are measured in presidential years, the 1869 newspaper data are treated as describing the 1868 newspaper market.
- **Newspaper-level variables.** For each paper-year, the authors record the paper's name, city, time of day, circulation, and political affiliation when available.
- **Matching over time.** Papers are matched across years using title, city, and time of day. Newspaper cities are mapped to Census places and then to counties.
- **Main county-level media measure.** For each county-year, the key variable is the number of daily newspapers headquartered in the county.

2.2 Readership measure

- **Goal.** The authors want a county-level estimate of the share of eligible voters reading at least one newspaper.
- **Circulation-to-readership conversion.** They assume each newspaper copy is read by two people, based on historical and recent readership evidence.
- **Overlap correction.** Because some people read multiple papers, they use a stylized overlap model: for any papers A and B, the share of A's readers who also read B equals the overall share reading B.
- **Use in interpretation.** Readership is not the main outcome; it is mainly used to translate the effect of newspaper entry into an implied effect of reading a newspaper.

2.3 Political affiliation and newspaper content

- **Historical affiliation.** Newspapers often declared themselves Republican, Democratic, or Independent.
- **Permanent classification.** A paper is classified as Republican if it ever declares a Republican affiliation, Democratic if it ever declares a Democratic affiliation, and by majority affiliation if it declares both in different years.
- **Final affiliation counts.** The sample includes 2,566 Republican papers, 2,431 Democratic papers, 1,714 Independent papers, and 1,063 papers that never report an affiliation.

- **Partisan balance measure.** For each county-year, the paper constructs the difference between the number of Republican and Democratic newspapers:

$$R_{ct} - D_{ct}.$$

  This is the key treatment for party vote-share regressions.

- **Nonpolitical newspapers.** The authors classify 307 papers as "nonpolitical" when they identify as commercial, financial, legal, trade, or another type unlikely to emphasize political news. These papers are useful as a placebo comparison.

- **Text-based content data.** For presidential elections from 1872 to 1928, the authors search newspaperarchive.com for articles mentioning presidential and vice-presidential candidates. The key content variable is the Republican share of candidate mentions.

- **Endorsement data.** For 1932–2004, the paper also uses *Editor and Publisher* presidential endorsement data.

2.4 Market definition

- **Baseline market.** The news market is defined as a county.
- **Why county?** County is the smallest geographic unit available consistently for presidential election data over the full historical period.
- **Measurement issue.** Some counties contain multiple local news markets, and some newspapers circulate outside their home county. This creates measurement error.
- **Why the approximation is reasonable.** Modern data suggest that the median newspaper sells more than 80 percent of its copies in its home county, and the median county with a newspaper receives more than 80 percent of its copies from in-county newspapers.
- **Robustness.** The paper checks results on isolated counties where the county-market approximation is likely to be more accurate.

2.5 Election and demographic data

- **Voting data.** The authors match counties to election data for president, House representative, governor, and senator through 2004.
- **Turnout.** Turnout is total votes divided by eligible voters. Eligible voters are constructed from Census demographic data.
- **Party vote share.** The main vote-share outcome is the Republican share of the two-party vote.
- **Incumbency outcomes.** For congressional races, the paper studies incumbent two-party vote share and whether the incumbent runs effectively uncontested.
- **Demographic controls.** The baseline controls include changes in:
  - share white,
  - share of whites who are foreign-born,
  - share male aged 21 and older,

- share living in cities with at least 25,000 residents,
- share living in towns with at least 2,500 residents,
- manufacturing employment relative to males 21 and older,
- log manufacturing output per capita, used as an income proxy.

2.6 Key descriptive facts

- **Entries and exits are common.** The paper observes thousands of net entry and exit events over the sample.
- **Events are large.** Entry and exit create large discrete changes in local newspaper availability. A typical newspaper enters at 87 percent of its lifetime average circulation and exits after having 115 percent of its lifetime average circulation in the previous year.
- **Average readership effect.** A typical newspaper event changes the share of eligible voters reading at least one newspaper by about 13 percentage points.
- **Main drivers of entry.** Population and income are the two most important market-level predictors of newspaper entry and exit. Population alone explains 61 percent of the cross-county variation in the average number of daily newspapers.
- **Political affiliation is meaningful.** Republican papers devote a larger share of candidate mentions to Republican candidates, and historically Republican papers are much more likely than historically Democratic papers to endorse Republican candidates later in the sample.

3 Empirical specifications (all equations + interpretation)

3.1 Baseline causal model

Let c index counties, s states, and t presidential election years. One time step equals four calendar years. The outcome y_{ct} can be turnout, Republican vote share, or incumbent vote share.

The paper begins with the following model:

$$y_{ct} = \rho_c + \beta n_{ct} + \gamma_{st} + \delta x_{ct} + \lambda z_{ct} + \varepsilon_{ct}. \tag{1}$$

Interpretation.

- n_{ct} is the number of newspapers in county c in year t.
- ρ_c is a county fixed effect.
- γ_{st} is a state-year effect. It absorbs election-specific statewide factors, such as whether the state is electorally competitive.
- x_{ct} is the vector of observable county demographics and income proxies.
- z_{ct} is unobserved newspaper profitability.
- β is the causal effect of newspaper availability on the political outcome.

- λz_{ct} is the main omitted-variable concern: newspaper profitability may be correlated with politics.

The core problem is that z_{ct} is not observed. Newspapers enter when profits are high, and profits are related to population, income, advertising demand, and possibly political conditions.

3.2 First-difference estimating equation

Because the data have persistent shocks, the paper estimates the model in first differences:

$$\Delta y_{ct} = \beta \Delta n_{ct} + \Delta \gamma_{st} + \delta \Delta x_{ct} + \lambda \Delta z_{ct} + \Delta \varepsilon_{ct}. \tag{2}$$

Interpretation.

- $\Delta y_{ct} = y_{ct} - y_{c,t-1}$ is the change in the outcome between presidential elections.
- Δn_{ct} is net newspaper entry or exit.
- County fixed effects drop out mechanically.
- State-year fixed effects mean the comparison is within the same state and election year.
- The estimate asks: when a county gains a newspaper, does its turnout or vote share change more than other counties in the same state-year?

Standard errors are clustered by county to allow serial correlation within counties.

3.3 Identification condition

The key identifying condition is:

$$\mathbb{E}\left(\Delta \varepsilon_{ct} \mid \Delta \gamma_{st}, \Delta x_{ct}, \Delta z_{ct}, \Delta n_{ct}\right) = 0. \tag{3}$$

The authors also assume that entry is positively related to contemporaneous profit shocks:

$$\operatorname{Cov}\left(\Delta z_{ct}, \Delta n_{ct} \mid \Delta \gamma_{st}, \Delta x_{ct}\right) > 0. \tag{1}$$

Interpretation.

- The cleanest case is $\lambda = 0$: after controls and state-year fixed effects, profitability shocks do not directly predict political outcomes.
- If $\lambda \neq 0$, the direction of bias depends on whether the profit drivers of newspaper entry are associated with rising or falling political outcomes.
- The paper does not simply assume $\lambda = 0$. It uses theory, event-study timing, and placebo tests to assess likely bias.

3.4 Event-study / pretrend specification

To study timing, the authors estimate specifications of the form:

$$\Delta z_{ct} = \sum_{k=-10}^{10} \alpha_k \Delta n_{c,t-k} + \Delta \gamma_{st} + \delta \Delta x_{ct} + \Delta \varepsilon_{ct}. \tag{4}$$

The same event-study logic is then applied to political outcomes:

$$\Delta y_{ct} = \sum_{k=-10}^{10} \widetilde{\alpha}_k \Delta n_{c,t-k} + \Delta\gamma_{st} + \delta\Delta x_{ct} + \Delta\varepsilon_{ct}. \tag{5}$$

Interpretation.

- The left side is a change in an outcome over a four-year period.
- $k = 0$ is the on-impact effect of a newspaper entry or exit.
- $k < 0$ shows whether the outcome was already trending before the event.
- $k > 0$ shows whether changes persist or continue after the event.
- Because the model is in first differences, a positive spike at $k = 0$ corresponds to a permanent positive shift in the level of the outcome.

3.5 Turnout specification

For turnout, the outcome is:

$$y_{ct} = \text{Turnout}_{ct} = \frac{\text{total votes}_{ct}}{\text{eligible voters}_{ct}}. \tag{2}$$

The baseline turnout regression is:

$$\Delta\text{Turnout}_{ct} = \beta\Delta n_{ct} + \Delta\gamma_{st} + \delta\Delta x_{ct} + u_{ct}. \tag{6}$$

Interpretation.

- β is the effect of gaining one additional daily newspaper on voter turnout.
- The main estimate is about 0.0034, meaning 0.34 percentage points.
- The paper estimates this for presidential turnout, congressional turnout in presidential years, and congressional turnout in off-year elections.

3.6 Political versus nonpolitical newspapers

The placebo-style specification separates newspapers likely to carry political information from those unlikely to do so:

$$\Delta\text{Turnout}_{ct} = \beta_P\Delta n^P_{ct} + \beta_N\Delta n^N_{ct} + \Delta\gamma_{st} + \delta\Delta x_{ct} + u_{ct}. \tag{7}$$

Interpretation.

- n^P_{ct} is the number of political newspapers.
- n^N_{ct} is the number of nonpolitical newspapers.
- If the turnout result is truly driven by political information, political papers should affect turnout more than nonpolitical papers.
- The results support this: political papers have a positive effect; nonpolitical papers do not.

3.7 Market-structure specification

To test whether competition matters, the paper replaces the number of newspapers with indicators for having at least one, at least two, and at least three newspapers:

$$\Delta\text{Turnout}_{ct} = \beta_1 \Delta \mathbf{1}\{N_{ct} \geq 1\} + \beta_2 \Delta \mathbf{1}\{N_{ct} \geq 2\} + \beta_3 \Delta \mathbf{1}\{N_{ct} \geq 3\} + \Delta\gamma_{st} + \delta \Delta x_{ct} + u_{ct}. \tag{8}$$

Interpretation.

- β_1 is the marginal effect of the first newspaper.
- β_2 is the marginal effect of the second newspaper.
- β_3 is the marginal effect of the third newspaper.
- The first newspaper is much more important than additional competitors.

The paper also tests ideological diversity with an indicator for having at least one Republican and one Democratic paper:

$$\Delta\text{Turnout}_{ct} = \beta_1 \Delta \mathbf{1}\{N_{ct} \geq 1\} + \beta_2 \Delta \mathbf{1}\{N_{ct} \geq 2\} + \beta_3 \Delta \mathbf{1}\{N_{ct} \geq 3\} + \eta \Delta \mathbf{1}\{R_{ct} \geq 1, D_{ct} \geq 1\} + \cdots . \tag{3}$$

The estimate of η is small and statistically insignificant.

3.8 Partisan persuasion specification

To test whether partisan newspapers move party vote shares, the outcome is the Republican share of the two-party vote:

$$y_{ct} = \frac{\text{Republican votes}_{ct}}{\text{Republican votes}_{ct} + \text{Democratic votes}_{ct}}. \tag{4}$$

The key independent variable is the partisan newspaper balance:

$$R_{ct} - D_{ct}, \tag{5}$$

where R_{ct} is the number of Republican newspapers and D_{ct} is the number of Democratic newspapers. The estimating equation is:

$$\Delta y_{ct} = \theta \Delta (R_{ct} - D_{ct}) + \Delta\gamma_{st} + \delta \Delta x_{ct} + u_{ct}. \tag{9}$$

Interpretation.

- $\theta > 0$ would mean Republican papers increase Republican vote share and Democratic papers decrease it.
- The paper finds $\theta \approx 0$ for presidential elections.
- This is striking because a change in partisan newspaper balance strongly changes the partisan balance of circulation.

3.9 Incumbency specification

For incumbency, the analysis is at the congressional district level. The outcomes are:

$$y_{dt} \in \{\text{incumbent two-party vote share}_{dt}, \mathbf{1}\{\text{incumbent effectively uncontested}_{dt}\}\} . \tag{6}$$

The market-structure specification is analogous to equation (8), but with congressional district outcomes:

$$\Delta y_{dt} = \beta_1 \Delta \mathbf{1}\{N_{dt} \geq 1\} + \beta_2 \Delta \mathbf{1}\{N_{dt} \geq 2\} + \beta_3 \Delta \mathbf{1}\{N_{dt} \geq 3\} + \Delta \gamma_{st} + \delta \Delta x_{dt} + u_{dt}. \tag{10}$$

Interpretation.

- Newspapers could help incumbents by increasing name recognition.
- They could hurt incumbents by informing voters about poor performance or limiting incumbent control over information.
- Empirically, the estimates are negative but imprecise, so the paper does not find clear incumbency effects.

3.10 Magnitude calculations

For a first newspaper in a county:

- effect on readership: about 0.25;
- effect on turnout: about 0.010.

The implied effect of reading a newspaper on voting is:

$$\frac{0.010}{0.25} = 0.040. \tag{7}$$

Interpretation. Reading a newspaper raises the probability of voting by about **4 percentage points** under the paper's intent-to-treat calculation.

The paper also computes a persuasion-rate style turnout effect. In a no-newspaper county, about 69 percent of eligible voters vote, so 31 percent are potential marginal voters. If 25 percent read the entering paper, then 7.7 percent of eligible voters are potential readers who could be induced to vote. A 1.0 percentage point turnout increase implies:

$$\frac{0.010}{0.077} \approx 0.128. \tag{8}$$

So the turnout persuasion rate is about **12.8 percent**.

4 Identification and interpretation (what makes the estimates credible)

4.1 Why entry and exit timing is useful

- **Discrete events.** Newspaper entry and exit are lumpy, irreversible, and create large changes in media availability.

- **Small contemporaneous trends relative to event size.** Circulation trends before or after entry/exit are much smaller than the jump caused by the event.
- **Regression-discontinuity intuition.** Entry occurs when profitability crosses a threshold. The exact crossing date creates a discrete change in treatment that is large relative to smooth underlying profit trends.
- **Not pure random assignment.** Entry is still endogenous, but the paper argues that the precise timing gives useful quasi-experimental variation.

4.2 Role of state-year fixed effects

- State-year fixed effects absorb all shocks common to counties in the same state and election year.
- This is important because presidential incentives, electoral competitiveness, ballot rules, and campaign intensity vary sharply across states and years.
- The identifying comparison is within a state-year: counties with newspaper entry/exit versus counties without such changes.

4.3 Role of first differences

- First differencing removes permanent county-level political differences.
- It also handles highly persistent shocks better than a simple county fixed-effect levels model.
- The estimating variation is changes in media market structure, not levels of newspaper density.

4.4 Pretrend logic

- If newspaper entry is caused by economic growth, political outcomes may already be trending before entry.
- The authors use event studies to check whether changes in outcomes occur before the entry/exit event.
- For turnout, entry is preceded by negative turnout trends, largely explained by demographics. That pattern works *against* finding a positive turnout effect.
- After demographic controls, turnout pretrends largely disappear while the on-impact turnout increase remains.

4.5 Bias direction for turnout

- Newspaper entry is associated with population and income growth.
- Population growth is typically associated with lower turnout, partly because movers and newer residents have weaker local ties.
- Therefore, if anything, omitted market growth should bias the turnout effect downward.
- The estimated positive turnout effect is therefore hard to explain purely with omitted population or income trends.

4.6 Bias direction for party vote shares

- Partisan newspaper affiliation is endogenous.
- Republican papers are more likely to enter Republican-leaning or increasingly Republican markets.
- This should bias the estimated effect of Republican papers on Republican vote share upward.
- The fact that the estimated effect is still approximately zero is therefore strong evidence against large persuasive effects of explicitly partisan newspapers.

4.7 Placebo and robustness evidence

- **Political versus nonpolitical newspapers.** Political newspapers raise turnout; nonpolitical newspapers do not, even though both are associated with local economic growth.
- **Other counties in the same state.** Newspaper entry in one county does not predict sizable turnout changes in other counties in the same state.
- **Large versus small papers.** Effects are larger for events involving larger papers, consistent with a media exposure mechanism.
- **Isolated counties.** Results are stronger where the county is a better approximation to the newspaper market.
- **Alternative specifications.** The results are robust to flexible event-time trends, alternative treatment definitions, truncating changes in the number of newspapers, and other checks reported in the appendix.

4.8 Remaining limitations

- **Entry is not randomly assigned.** The strategy is credible because of timing, pretrends, controls, and placebo tests, not because entry is literally random.
- **Market definition is imperfect.** Counties are not always true newspaper markets, and papers can circulate across county boundaries.
- **Partisanship measure is explicit.** The vote-share results apply to declared Republican/Democratic affiliations. They may not generalize to subtle or hidden media bias.
- **Average historical effects.** The estimates average over many counties, years, and events. Particular newspapers may have had much larger or smaller effects.
- **Welfare is not directly measured.** Higher turnout is not automatically welfare-improving, and the paper does not directly evaluate policy quality or voter information accuracy.

5 Tables and figures

5.1 Table 1: Partisan affiliation and newspaper content

Read: Table 1 validates the paper's use of declared newspaper affiliation as a measure of political content.

- Republican-affiliated newspapers devote about **18.5 percentage points** more of their candidate mentions to Republican candidates than Democratic-affiliated newspapers.
- The relationship remains large after controlling for the Republican vote share in the local market.
- Even newspapers that later call themselves Independent retain content differences predicted by their historical affiliation.
- Historically Republican papers endorsed Republican presidential candidates 90 percent of the time during 1932–2004, compared with 45 percent for historically Democratic papers.

Why it matters. The partisan newspaper variable is not just a label. It captures meaningful differences in political content and endorsements.

TABLE 1—PARTISAN AFFILIATION AND NEWSPAPER CONTENT

	Republican share of candidate mentions			
	(1)	(2)	(3)	(4)
Republican affiliation (permanent)	0.1850 (0.0283)	0.1571 (0.0324)	0.1815 (0.0298)	0.1829 (0.1005)
Republican vote share		0.1858 (0.0850)		
Constant	0.2943 (0.0168)	0.2135 (0.0359)	0.2972 (0.0185)	0.2900 (0.0635)
Sample	All	All	Currently Independent? No	 Yes
Newspaper-years	423	423	363	60
R^2	0.2517	0.2591	0.2617	0.4482

Notes: Standard errors in parentheses are clustered by newspaper. Time period is 1872–1928. All specifications include year fixed effects. Each observation is a newspaper-year. Dependent variable is the number of search hits for the Republican presidential and vice presidential candidate divided by the total number of search hits for both Republican and Democratic candidates.

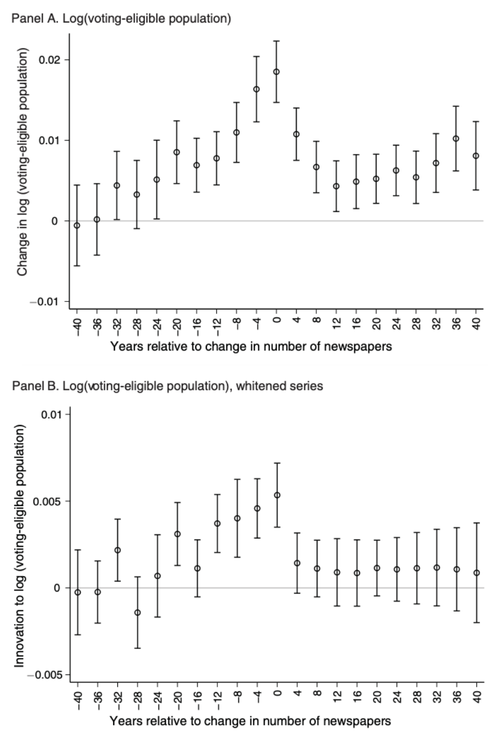

FIGURE 1. CHANGES IN POPULATION AROUND NEWSPAPER ENTRIES/EXITS

Notes: Panel A shows coefficients from a regression of change in log(voting-eligible population) on a vector of leads and lags of the change in the number of newspapers (see equation (4) for details). Panel B shows coefficients from a regression of innovation in log(voting-eligible population) on a vector of leads and lags of the change in the number of newspapers (see equation (4) for details). Innovation in a variable is its residual from a regression of the variable on ten lags of the variable and of the event indicator. Models include state-year fixed effects. Error bars are ±2 standard errors. Standard errors are clustered by county. Time period is 1868–1928.

5.2 Figure 1: Changes in population around newspaper entries/exits

Read: Figure 1 plots population changes before and after newspaper entry/exit events.

- Population growth is above average in the period of newspaper entry.
- Population growth is also above average before entry, showing that newspapers enter growing markets.
- Because population growth is serially correlated, growth also tends to remain high after entry.
- When the authors use a whitened population innovation, the postevent pattern disappears by construction.

Why it matters. This figure shows that entry is endogenous to local market growth, but it also shows how pretrends can diagnose the likely direction and magnitude of bias.

5.3 Figure 2: Presidential turnout around newspaper entries/exits

Read: Figure 2 is the central visual evidence for the turnout result.

- Without demographic controls, turnout shows a positive spike at the time of entry/exit of about 0.2 percentage points.
- There is a negative turnout pretrend before entry, consistent with growing areas having lower turnout.
- With demographic controls, the pretrend largely disappears.
- The on-impact effect rises to roughly 0.3 percentage points.

Why it matters. The effect appears exactly when a newspaper enters or exits, rather than gradually before the event. This supports a causal interpretation.

5.4 Table 2: Effect of newspaper entry/exit on voter turnout

Read: Table 2 reports the main regression estimates for readership and turnout in 1868–1928.

- A one-newspaper event raises readership by **0.1314**, or 13.14 percentage points.
- Without demographic controls, the presidential turnout effect is **0.0026**, or 0.26 percentage points.
- With demographic controls, the presidential turnout effect is **0.0034**, or 0.34 percentage points.
- The effect on congressional turnout in presidential years is **0.0031**.
- The effect on congressional turnout in off-years is **0.0032**.

Why it matters. This is the core quantitative result: newspaper entry/exit robustly changes voter participation, and the effect is similar across presidential and congressional elections.

5.5 Table 3: Political versus nonpolitical newspapers

Read: Table 3 separates political papers from papers unlikely to carry political news.

- Political newspapers are associated with higher turnout: coefficient **0.0037**.

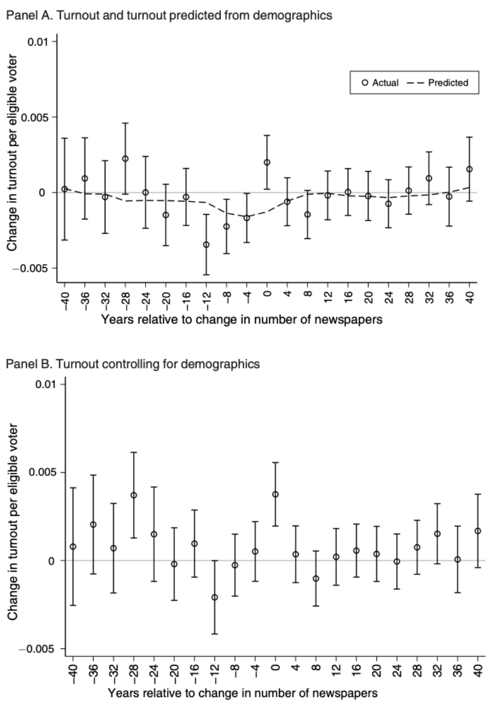

Figure 2. Presidential Turnout and Newspaper Entries/Exits

Notes: Panel A shows coefficients from a regression of change in turnout per eligible voter on a vector of leads and lags of the change in the number of newspapers (see equation (4) for details). "Actual" refers to estimated coefficients. "Predicted" refers to estimated coefficients using as a dependent measure the change in turnout predicted from demographics as defined in Section ID. Panel B shows coefficients from a regression of change in turnout per eligible voter, controlling for demographics, on a vector of leads and lags of the change in the number of newspapers. Models include state-year fixed effects. Error bars are ±2 standard errors. Standard errors are clustered by county. Time period is 1868–1928.

Table 2—The Effect of Newspaper Entry/Exit on Voter Turnout

	Readership	Presidential turnout		Congressional turnout	
				(Pres. years)	(Off years)
	(1)	(2)	(3)	(4)	(5)
Number of newspapers	0.1314 (0.0044)	0.0026 (0.0009)	0.0034 (0.0009)	0.0031 (0.0011)	0.0032 (0.0012)
Demographic controls?	Yes	No	Yes	Yes	Yes
R^2	0.435	0.569	0.579	0.521	0.531
Number of counties	1,181	1,195	1,195	1,195	1,192
Number of county-years	11,281	15,627	15,627	14,634	13,869

Notes: Standard errors in parentheses are clustered by county. Time period is 1868–1928. Models are estimated in first differences. All specifications include state-year fixed effects. Demographic controls are changes in county demographics as defined in Section ID, with dummies included for missing data. The dependent variable in column 4 is the change in congressional turnout between the current presidential year (t) and the previous presidential year ($t-4$). The dependent variable in column 5 is the "long difference": congressional turnout in the first off-year election following the current presidential election year ($t+2$) minus congressional turnout in the last off-year election prior to the previous presidential election year ($t-6$).

- Nonpolitical newspapers have a negative and statistically insignificant coefficient: −0.0022.
- Both types are positively related to population growth, so the difference is unlikely to be explained simply by growing markets.
- The equality test for turnout has $p = 0.0567$, meaning the difference is close to conventional significance.

Why it matters. The turnout effect appears specific to newspapers that plausibly provide political information.

TABLE 3— TURNOUT EFFECTS OF POLITICAL AND NONPOLITICAL NEWSPAPERS

	Log(voting-eligible population) (1)	Presidential turnout (2)
Number of political newspapers	0.0072	0.0037
	(0.0013)	(0.0010)
Number of nonpolitical newspapers	0.0137	−0.0022
	(0.0046)	(0.0028)
F-test of equality of coefficients	2.122	3.639
p-value	0.1454	0.0567
R^2	0.816	0.579
Number of counties	1,195	1,195
Number of county-years	15,627	15,627

Notes: Standard errors in parentheses are clustered by county. Time period is 1868–1928. Models are estimated in first differences. All specifications include state-year fixed effects and demographic controls as defined in Section ID, with dummies included for missing data. Nonpolitical newspapers are those that identify themselves as commercial, financial, legal, trade, or other types of publications unlikely to emphasize political news. All other newspapers are classified as political. See Section I for details.

TABLE 4—TURNOUT EFFECTS BY NUMBER OF NEWSPAPERS AND IDEOLOGICAL DIVERSITY

	Readership	Presidential turnout	
	(1)	(2)	(3)
County has:			
≥ 1 newspaper	0.2470	0.0098	0.0121
	(0.0082)	(0.0027)	(0.0033)
≥ 2 newspapers	0.1030	−0.0018	−0.0050
	(0.0068)	(0.0022)	(0.0032)
≥ 3 newspapers	0.0710	0.0052	0.0030
	(0.0067)	(0.0024)	(0.0034)
At least one Republican and one Democratic paper			−0.0023
			(0.0040)
F-test of equality of coefficients	146.8	5.949	—
p-value	0.0000	0.0027	—
R^2	0.500	0.579	0.578
Number of counties	1,181	1,195	1,168
Number of county-years	11,281	15,627	12,515

Notes: Standard errors in parentheses are clustered by county. Time period is 1868–1928. Models are estimated in first differences. All specifications include state-year fixed effects and demographic controls as defined in Section ID, with dummies included for missing data.

5.6 Table 4: Turnout effects by number of newspapers and ideological diversity

Read: Table 4 asks whether the first newspaper matters more than additional competitors.

- The first newspaper increases readership by **0.2470** and presidential turnout by **0.0098**.
- The marginal effect of a second newspaper on turnout is −0.0018, not statistically significant.
- The marginal effect of a third newspaper is **0.0052**, smaller and less stable than the first-paper effect.
- The additional effect of having both a Republican and a Democratic paper is −0.0023, statistically insignificant.
- The equality test rejects equal turnout effects across first, second, and third newspapers.

Why it matters. Newspaper competition is not the main driver. The main effect comes from moving from no local daily paper to at least one paper.

5.7 Table 5: Turnout effects over time

Read: Table 5 estimates different effects in the newspaper, radio, and television eras.

- For presidential turnout, the first newspaper effect is **0.0084** in 1868–1928.
- It falls to **0.0054** in the radio period, 1932–1952.
- It is nearly zero, **0.0010**, in the television period, 1956–2004.

- For congressional turnout, the estimates remain positive in later periods, especially for off-year elections.
- Readership effects grow over time because later entering/exiting papers tend to have larger circulation.

Why it matters. The decline for presidential turnout is consistent with substitution toward radio and television for national political information. The persistence for congressional turnout fits the idea that newspapers remain more important for local and lower-salience politics.

TABLE 5—TURNOUT EFFECTS OVER TIME

	Readership (1)	Presidential turnout (2)	Congressional turnout (pres. years) (3)	Congressional turnout (off years) (4)
County has $\geq$ 1 newspaper:				
Newspaper period	0.2591	0.0084	0.0070	0.0110
(1868–1928)	(0.0084)	(0.0027)	(0.0030)	(0.0031)
Radio period	0.4512	0.0054	0.0042	0.0051
(1932–1952)	(0.0191)	(0.0032)	(0.0042)	(0.0043)
Television period	0.4681	0.0010	0.0064	0.0070
(1956–2004)	(0.0161)	(0.0021)	(0.0054)	(0.0042)
F-test of equality of coefficients	90.87	2.466	0.151	0.711
p-value	0.0000	0.0852	0.860	0.491
R^2	0.492	0.602	0.497	0.602
Number of counties	1,489	1,489	1,486	1,486
Number of county-years	41,840	46,899	38,130	37,339

Notes: Standard errors in parentheses are clustered by county. Time period is 1868–2004 for presidential turnout and 1868–1990 for congressional turnout. Models are estimated in first differences. All specifications include state-year fixed effects and demographic controls as defined in Section ID, with dummies included for missing data. The dependent variable in column 3 is the change in congressional turnout between the current presidential year (t) and the previous presidential year ($t - 4$). The dependent variable in column 4 is the "long difference": congressional turnout in the first off-year election following the current presidential election year ($t + 2$) minus congressional turnout in the last off-year election prior to the previous presidential election year ($t - 6$).

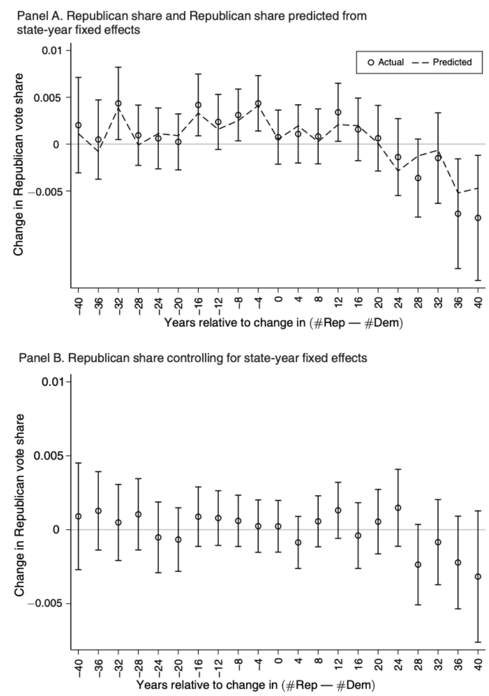

FIGURE 3. REPUBLICAN VOTE SHARE AND NEWSPAPER ENTRIES/EXITS

Notes: Panel A shows coefficients from a regression of change in Republican share of two-party vote on a vector of leads and lags of changes in the difference in the number of Republican and Democratic newspapers (see equation (4) for details). "Actual" refers to estimated coefficients. "Predicted" refers to estimated coefficients using as a dependent measure the change in Republican share of two-party vote predicted from state-year fixed effects alone. Panel B shows coefficients from a regression of change in Republican share of two-party vote, controlling for state-year fixed effects, on a vector of leads and lags of changes in the difference in the number of Republican and Democratic newspapers. Error bars are ± 2 standard errors. Standard errors are clustered by county. Time period is 1868–1928.

5.8 Figure 3: Republican vote share around partisan newspaper entries/exits

Read: Figure 3 plots changes in Republican vote share around changes in the Republican-minus-Democratic newspaper balance.

- Without controls, Republican papers tend to enter markets where Republican vote share is already trending upward.
- There is no sharp on-impact jump in Republican vote share at the event date.
- With state-year fixed effects, the broad trend largely disappears.
- The on-impact effect remains statistically indistinguishable from zero.

Why it matters. The figure directly addresses persuasion. Partisan paper entry does not produce the sharp vote-share response one would expect from strong persuasion.

5.9 Table 6: Effect of partisan newspaper entry/exit on Republican vote share

Read: Table 6 quantifies the partisan persuasion result.

- A change in the Republican-minus-Democratic newspaper count shifts the Republican-minus-Democratic share of circulation by **0.5096**.
- The effect on Republican presidential vote share is only **0.0002**, with or without demographic controls.
- The effect on Republican congressional vote share is **0.0021**, statistically insignificant.

Why it matters. Partisan newspaper entry changes what people could read, but it does not measurably change which party receives votes.

TABLE 6—THE EFFECT OF PARTISAN NEWSPAPER ENTRY/EXIT ON REPUBLICAN VOTE SHARE

	Circulation Rep share – Dem share (1)	Presidential vote share (2)	Presidential vote share (3)	Congressional vote share (4)
(# Rep – # Dem) newspapers	0.5096	0.0002	0.0002	0.0021
	(0.0135)	(0.0010)	(0.0010)	(0.0018)
Demographic controls?	yes	no	yes	yes
R^2	0.586	0.735	0.736	0.351
Number of counties	1,181	1,195	1,195	1,191
Number of county-years	11,281	15,401	15,401	14,295

Notes: Standard errors in parentheses are clustered by county. Time period is 1868–1928. Models are estimated in first differences. All specifications include state-year fixed effects. Demographic controls are changes in county demographics as defined in Section ID, with dummies included for missing data.

TABLE 7— INCUMBENCY EFFECTS BY NUMBER OF NEWSPAPERS

	Congressional incumbent vote share (1)	Uncontested incumbent (2)
Effect of having:		
$\geq$ 1 newspaper	−0.0170	−0.0432
	(0.0183)	(0.0597)
$\geq$ 2 newspapers	0.0019	−0.0139
	(0.0112)	(0.0339)
$\geq$ 3 newspapers	−0.0136	−0.0193
	(0.0112)	(0.0285)
R^2	0.583	0.388
Number of districts	319	355
Number of district-years	901	1206

Notes: Standard errors in parentheses are clustered by district. Time period is 1868–1928. Models are estimated in first differences. All specifications include state-year fixed effects and demographic controls as defined in Section ID, with dummies included for missing data. Congressional incumbent vote share is the share of the two-party vote received by the incumbent candidate. Uncontested incumbent is a dummy for whether the incumbent candidate received more than 95 percent of the total vote. Sample excludes district-years in which there is no incumbent in the current or previous presidential election year, the incumbent in the current or previous presidential election year was neither Republican nor Democratic, or there was a redistricting in the state between the current and previous presidential election year.

5.10 Table 7: Incumbency effects by number of newspapers

Read: Table 7 tests whether newspapers affect incumbency advantage.

- The effect of having at least one newspaper on the incumbent's congressional vote share is -0.0170, but it is statistically insignificant.
- The effect on the probability the incumbent is effectively uncontested is -0.0432, also statistically insignificant.
- Effects of second and third newspapers are also statistically insignificant.

Why it matters. The evidence does not support a clear average effect of newspapers on incumbency advantage. Newspapers may help some incumbents and hurt others, but there is no systematic average pattern.

5.11 Appendix robustness checks

Read: The appendix shows that the main results are not driven by a narrow specification choice.

- Turnout effects are robust to alternative definitions of newspaper events.
- Estimates are larger for larger newspapers and in isolated counties.
- Results survive flexible controls for event-time trends.
- Vote-share null effects remain robust across alternative definitions of partisanship and samples.

Why it matters. The central pattern is stable: newspapers mobilize voters, but explicitly partisan newspapers do not generate large party vote-share shifts.

6 Short summary

- **Method.** Build a long historical panel of U.S. daily newspapers and county election outcomes; estimate first-difference models using entry/exit timing, state-year fixed effects, controls, and event-study pretrends.
- **Main turnout result.** One additional newspaper raises presidential and congressional turnout by about 0.3 percentage points. The first newspaper in a county raises turnout by about 1 percentage point.
- **Mechanism.** The effect is concentrated in political newspapers and is proportional to readership. This supports an information/mobilization channel.
- **Competition result.** The first paper matters much more than the second or third. Competition and ideological diversity do not appear to be the key drivers of turnout.
- **Persuasion result.** Declared partisan newspapers do not measurably shift party vote shares, despite meaningful differences in content and endorsements.
- **Incumbency result.** There is no clear evidence that newspapers systematically help or hurt incumbents.
- **Interpretation.** Newspapers seem to make people more likely to vote by giving them political information, but they do not strongly persuade voters to support a particular party on average.

7 Conclusion

7.1 Core conclusion

Local newspapers have a measurable causal effect on political participation. The strongest evidence is that entry or exit of newspapers produces sharp changes in turnout, especially when a county moves from no newspaper to one newspaper.

7.2 Economic intuition

The first newspaper in a market substantially increases access to political information. Voters who previously had little local political information become more likely to participate. Additional newspapers provide less new information because many voters already have access to at least one paper. Declared partisan newspapers may change the slant of available content, but voters either select into ideologically preferred papers, discount obvious partisan bias, or are not easily persuaded across party lines.

7.3 Takeaway

The paper's central message is not that media always persuades voters. It is that media supply can change whether citizens participate in politics. Newspapers matter most when they fill an information vacuum; once some political information is already available, additional competition or explicit partisanship has much smaller average effects.